

Mini-Project-1: AI Agent Analysis and Probabilistic Modeling-Theory

Team project

- Submission deadline: February 5th at 11:59
- Presentation deadline February 5th at class time.
- Every time, a member should be ready to present and answer questions. No presentation, no grade.
- Grades will be **individual based on the presentation and the domain of the topic.**

Paper: A Platform-Based Incentive Mechanism for Autonomous Vehicle Crowdsensing (Chakeri and Jaimes, IEEE OJ-ITS 2021)

Learning Objectives

- By completing this assignment, students will be able to classify real research systems using AI agent models, analyze task environment properties, interpret utility-based decision making, understand probabilistic modeling, and examine algorithmic complexity.

Part 1 – Agent Type (MAI Classification)	1
Part 2 – Task Environment Classification.....	3
Part 3 – Knowledge Representation	3
Part 4 – Numerical Reasoning	5
Part 5 – Probability Modeling.....	6
Part 6 – Combinatorial Complexity	7
Part 7 – Conceptual AI Links	8

Part 1 – Agent Type (MAI Classification)

1. According to Russell and Norvig (MAI), what type(s) of agent are the Autonomous Vehicles (AVs)? Mark all that applies and justify your answer with evidence from the paper.

- Simple reflex agent:
- **Model-based agent: The agent has a model of the entire world, covering all the roads and destinations.**
- **Goal-based agent: The agent has a goal of providing the most amount of money to each driver.**
- **Utility-based agent: The agent has access to the internal state of the car; it has the distance, amount of work and money made.**
- Learning agent:

2. Why is this system not a simple reflex agent?

The system is too complex for a simple reflex agent. Simple reflex agents do not keep a

memory of events.

3. Why is this system not a learning agent?

The system cannot change its algorithm/decision table while it is running.

4. Explain why this system is better classified as a utility-based agent rather than purely goal-based.

A utility-based agent is factors in the current state and tries to achieve best optimizations, while a goal-based agent needs to think and plan about the future and takes actions in multiple steps. The AV agents are

5. What metrics or measurement criteria are used to evaluate whether the agent is performing well or poorly

- Amount of utility that a car can achieve
- The amount of budget/time per utility
- The performance when there are a lot of vehicles on the road
- Reducing complexity for getting the optimal solution
- Reducing uncertainty and inconsistency with results
- Road coverage and crowdsources coverage
- How well it does on incomplete and incomplete knowledge

6. Are these agents rational?

A rational agent chooses the best action based on the history and current circumstances, which applies to the Av agent

7. Is the resulting set of trajectories optimal, meaning that they represent the best trajectories that can be obtained?

Based on the given result of the other models, such as the greedy and random, the agent does far better than the rest

Part 2 – Task Environment Classification

For Autonomous Vehicles in this system, classify the task environment and justify your answers.

1. **Observability (Fully or Partially observable):** Both experiments had full information on vehicles, while a different experiment tested partial info. However, in the real world where this will be applied, I would go with partial observability, as an application won't have all cars, dialed in to factor.
2. **Deterministic or Stochastic:** It is Stochastic as it makes its best guess to have the best action in the moment; it cannot predict the future.
3. **Episodic or Sequential:** It is sequential as actions performed have long lasting consequences. A change in the trajectory can have a big impact on the utility/coverage.
4. **Static or Dynamic:** In the system, it is dynamic, as each vehicle can have a limited time to query knowledge about its other vehicles. During that time, the environment will change.
5. **Discrete or Continuous:** The environment is discrete, as the experiments are limited by the number of tasks/locations.
6. **Single-agent or multi-agent:** The environment is absolutely multi-agent; it has hundreds of vehicles competing against each other.
7. **Cooperative or Competitive:** It is a cooperative agent; each agent will work with each other to find an equilibrium to maximize a team's reward.

5. Are the AVs autonomous agents according to MAI's definition of autonomy? Explain.

The AV is not autonomous even though it does not need human intervention; they are goal-oriented and can make decisions. They don't have the ability to learn or modify its decision table.

Part 3 – Knowledge Representation

6. What internal knowledge must an AV maintain to make decisions?

The internal knowledge maintained by the autonomous vehicle must include the starting/ending location and the shortest path between them, the amount of data the AV can collect through its sensors, the information from the valid adjacent cells and crowdsourcers, and the Memory Buffer to avoid returning to already-visited places.

8. How does the Expected Utility equation guide decision making?

By calculating the expected utility, the AV can decide which of the available neighboring regions it can travel to that contains the highest value, maximizing the profit on its path.

9. How does the Expected Sensing Plan affect movement decisions?

The Expected Sensing Plan can impact the AV's route by determining whether cells with the highest expected utility are worth traveling to based on the AV's sensing capacity being greater than or equal to the calculated ESP for that cell. If the AV cannot accommodate the ESP of a neighboring cell in its capacity, it chooses another location where it can collect data.

10. What is the expected result, or what are the characteristics of the set of vehicles' trajectories? (from source to destination, plus the deviation to visit some sensing-interest (PoI) locations).

The characteristics of the AV's trajectories are the Approximated Trajectory Nash Equilibrium, where it maintains a stable configuration and cannot gain any profit by changing its course; deviating from the shortest path to visit areas not on the original path to collect data; and heading straight for its final destination after reaching its sensing capacity.

Part 4 – Numerical Reasoning

11. Expected Utility: An AV (me AV1) considers moving to a cell with reward $R = 120$. Two other AVs may also go there with probabilities of 0.6 and 0.3, respectively. Compute the expected utility of AV1 to visit the cells that contain R . Assume that if AV1 arrives there alone, its utility will be 120 (the whole reward)

To find the expected utility we need to find all possible scenarios where AV1 visits the cell. Likelihoods:

AV1 is alone: $(1-0.6)*(1-0.3) = 0.28$

AV1 with AV2 only: $0.6*(1-0.3) = 0.42$

AV1 with AV3 only: $(1-0.6)*0.3 = 0.12$

All three AVs: $0.6*0.3 = 0.18$

The reward R is decided by the number of agents present in each scenario

AV1 alone: $120/1 = 120$

AV1 with AV2: $120/2 = 60$

AV1 with AV3: $120/2 = 60$

All three AVs: $120/3 = 40$

The expected utility is the sum of the utility of each outcome multiplied by its probability

$$EU = (0.28 * 120) + (0.42 * 60) + (0.12 * 60) + (0.18 * 40) = 73.2$$

The expected utility for AV1 is 73.2

12. Capacity Constraint: An AV has sensing capacity $b = 8$. Expected sensing plans for upcoming cells are [3, 4, 5]. Which cells can it visit before exhausting capacity?

sensing capacity = 8

Cell 1: 3 units ($8 - 3 = 5$)

Cell 2: 4 units ($5 - 4 = 1$)

Cell 3: 5 units (the AV cannot complete the third cell because the 5 units > 1 remaining sensing capacity.)

Therefore the AV cannot complete visit the first two cells before exhausting capacity.

13. Sensing Game Intuition: If more AVs arrive at a crowdsourcer, how does that affect (a) individual sensing plan and (b) individual utility? Explain conceptually.

a) as more AVs arrive at the same crowdsourcer each AV usually contributes less sensing effort because the total required sensings is spread across more participants.

b) individual utility generally decreases as more AVs arrive because the reward is divided among more agents.

While the work per AV may decrease so will the reward per AV.

Part 5 – Probability Modeling

14. Binomial Distribution: If there are $n = 5$ nearby vehicles and each independently moves to a cell with probability $p = 0.4$, what is the probability exactly 2 of them go there?

$$P(X = 2) = \binom{5}{2} (0.4)^2 (0.6)^{5-2}$$

$$\binom{5}{2} = \frac{5!}{2!3!} = 10$$

$$P(X = 2) = 10 * 0.16 * 0.216 = \mathbf{0.3456}$$

15. Explain why the situation in the paper does not strictly follow a Binomial distribution. What distribution is more accurate?

This does not follow binomial distribution because there isn't a simple "choose to opt in or opt out". Instead it follows Poisson-Binomial distribution, since there are several different agents who all have different probabilities of participating.

14. The paper uses a Poisson-Binomial approximation. Explain: (a) what makes the distribution Poisson-Binomial, (b) why this reduces computational complexity, (c) why such an approximation is useful in multi-agent settings.

It's Poisson-Binomial because each instance has different probabilities. This reduces computational complexity because it allows us to generalize the calculation of all the different instances together, reducing the time needed to calculate it. This style of computation works well with multi-agent settings because it would allow for the rapid calculation of probabilities, even with thousands of agents being present in the system.

Part 6 – Combinatorial Complexity

15. Identify the main combinatorial explosion problem in this paper.

The problem discussed in the paper uses multiple crowdsourcers where their $C = \{c_1, c_2, c_F\}$ and participants $V = \{v_1, v_2, v_M\}$, each with a starting point, $S = \{s_1, s_2, s_M\}$, and a destination, $D = \{d_1, d_2, d_M\}$. Each crowdsourcer has a reward and destination sub-region. As the number of crowdsourcers goes up, the number of strategy combinations also increases, and as such it complicates the search quickly.

16. Why is finding the exact Nash equilibrium intractable? Discuss number of agents, trajectories, and strategy combinations.

Finding the exact Nash equilibrium is intractable because of the computational complexity. As the number of agents goes up or the number of trajectories that an agent can take go up, the number of different strategy combinations follows suit and with that it would take too long to find one for every situation.

17. How does the paper reduce complexity using a greedy local decision rule and approximated trajectory Nash equilibrium (ATNE)?

With the greedy local decision rule, the agents are only looking at their nearby area, which means it only focuses on the rewards and neighbors within the neighborhood of the participant instead of the whole picture which helps to cut down on the time to compute and with the approximated Nash equilibrium the agent is calculating the movements that are made based on the locations of those neighbors only. Then it adjusts based on the actual movements made.

18. Why does computing expected utility exactly lead to exponential complexity? How does the Poisson-Binomial approximation help?

Computing expected utility leads to exponential complexity due to the fact that there are many factors and possible things to calculate. The Poisson-Binomial approximation helps by reducing the complexity as every participant will have the same cost.

Part 7 – Conceptual AI Links

19. How is this system like a multi-agent rational agent system in MAI?

It is rational as each individual autonomous vehicle is a greedy agent that achieves maximum utility. They are all autonomous, and goal oriented for themselves instead of a group. Each agent makes an individual decision to maximize their utility in a competitive environment.

20. If Reinforcement Learning replaced the greedy expected-utility approach, how would the agent's classification change?

The agent's classification would change to a learning agent as it will be able to change the approach based on feedback it learned from the system. If new rules were applied to the game, it would be able to adapt. The greedy agent cannot change its approach.